

Working with Signed Rational Numbers

Answer Key

Grades 6–7 The Number System

Quick Answers

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|---|---|
| 1. -5 | 7. -4.75 |
| 2. 5 | 8. — |
| 3. -28 | 9. -11 |
| 4. -1 , $-\frac{3}{4}$, -0.6 , $\frac{1}{2}$ | 10. -8 |
| 5. -11 | 11. 7 |
| 6. $-\frac{2}{3}$ | 12. -2.05 , $-\frac{9}{4}$, -2.5 , $-\frac{11}{4}$ |
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Curriculum

Level: Grades 6–7 The Number System

6.NS.C / 7.NS.A – *problems 4, 8, 12*

7.NS.A.1, 7.NS.A.2 – *problems 1, 2, 3, 5, 6, 7, 9, 10, 11*

Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

10. *If they got -2: multiplied before dividing instead of working left to right*

1. Evaluate $-8 + 3$.

Step 1. The signs differ, so the two numbers pull in opposite directions: start at -8 and move 3 units toward zero.

Step 2. -8 is 8 units from zero and 3 is only 3, so the answer keeps the negative sign, and the size is $8 - 3 = 5$.

-5

Listen for: -11 , which comes from adding the sizes instead of subtracting them.

2. Evaluate $-6 - (-11)$.

Step 1. Subtracting a number is adding its opposite, so rewrite first: $-6 - (-11) = -6 + 11$.

Step 2. Now the signs differ: 11 is further from zero than 6, so the answer is positive, with size $11 - 6 = 5$.

5

Listen for: -17 — the two minus signs were read as one.

3. Evaluate $(-4)(7)$.**Step 1.** Multiply the sizes, then decide the sign by counting negative factors.**Step 2.** $4 \times 7 = 28$, and there is exactly one negative factor — an odd count — so the product is negative.

$$\boxed{-28}$$

Teaching note: four groups of 7 removed is the story behind the rule.**4. Order -1 , $-\frac{3}{4}$, -0.6 , $\frac{1}{2}$ from least to greatest.****Step 1.** The numbers are written three different ways, so convert them to one form — decimals: -1 , -0.75 , -0.6 , 0.5 .**Step 2.** On a number line, least is furthest left. All three negatives come before the positive, and among the negatives the one furthest from zero, -1 , is smallest.

$$\boxed{-1 < -\frac{3}{4} < -0.6 < \frac{1}{2}}$$

Listen for: $-0.6 < -0.75$, from ordering the negatives as if they were positive.**5. Evaluate $-3 + 4(-2)$.****Step 1.** Multiplication outranks addition, so $4(-2)$ must be evaluated before anything is added.**Step 2.** $4(-2) = -8$, so the expression is $-3 + (-8)$.**Step 3.** Two negatives added: the sizes add and the sign stays negative, giving -11 .

$$\boxed{-11}$$

Listen for: -2 , from adding $-3 + 4$ first.**6. Evaluate $(-\frac{3}{4})(\frac{8}{9})$.****Step 1.** One negative factor, so the answer is negative; multiply the sizes as fractions.**Step 2.** Multiply straight across: $\frac{3 \times 8}{4 \times 9} = \frac{24}{36}$.**Step 3.** $\frac{24}{36} = \frac{2}{3}$ in lowest terms, and the sign is negative.

$$\boxed{-\frac{2}{3}}$$

Faster route: cancel before multiplying — 8 and 4 share a factor of 4, and 3 and 9 share a factor of 3.**7. Evaluate $2.5 + (-7.25)$.****Step 1.** Different signs again, so subtract the smaller size from the larger and keep the sign of the larger.**Step 2.** $7.25 - 2.5 = 4.75$, and -7.25 is the one further from zero, so the answer is negative.

$$\boxed{-4.75}$$

Check: lining the decimal points up matters here — $7.25 - 2.5$ is not $7.25 - 2.05$.**8. Compare $-\frac{5}{6}$ and -0.8 .****Step 1.** One is a fraction and one is a decimal, so put them in the same form before comparing: $-\frac{5}{6} = -0.8333\dots$ **Step 2.** Both are negative, so the one further from zero is smaller. $0.8333\dots > 0.8$, so $-\frac{5}{6}$ sits further left.

$$\boxed{-\frac{5}{6} < -0.8}$$

Listen for: the reversed answer, from comparing 0.8333 with 0.8 and forgetting that both are negative.

9. Evaluate $(-2)^3 + 12 \div (-4)$.

Step 1. Exponents come first, then division, and only then the addition.

Step 2. $(-2)^3 = (-2)(-2)(-2) = -8$ — an odd number of negative factors, so the result is negative.

Step 3. $12 \div (-4) = -3$, so the expression is $-8 + (-3) = -11$.

-11

Listen for: 8 for $(-2)^3$; that is $(-2)^2 \cdot 2$ thinking. Note also that $(-2)^3$ and -2^3 agree here, while $(-2)^2$ and -2^2 do not.

10. Priya's $-24 \div (-6) \cdot (-2)$.

Step 1. Division and multiplication have equal rank, so they run *left to right* — Priya multiplied $(-6)(-2)$ first, which changes the meaning of the expression.

Step 2. Left to right: $-24 \div (-6) = 4$ (two negatives, so positive).

Step 3. Then $4 \cdot (-2) = -8$.

-8

What Priya did wrong: she computed $-24 \div [(-6)(-2)] = -24 \div 12 = -2$. Equal-rank operations are never reordered.

11. Evaluate $-3(2 - 5) - \frac{4^2}{8}$.

Step 1. Brackets first, then the exponent and the division, then the subtraction that joins the two halves.

Step 2. $2 - 5 = -3$, so the first term is $-3(-3) = 9$ — two negative factors, so positive.

Step 3. $4^2 = 16$ and $16 \div 8 = 2$, so the expression is $9 - 2$.

7

Listen for: -11, from multiplying $-3 \cdot 2$ before doing the bracket.

12. Challenge: order -2.5 , $-\frac{11}{4}$, -2.05 , $-\frac{9}{4}$ **from greatest to least.**

Step 1. Four negatives written two ways — convert the fractions to decimals so a single comparison rule applies: $-\frac{11}{4} = -2.75$ and $-\frac{9}{4} = -2.25$.

Step 2. The list is now -2.5 , -2.75 , -2.05 , -2.25 . Greatest means closest to zero, so read them by distance from zero: 2.05, 2.25, 2.5, 2.75.

Step 3. Greatest to least reverses that size order, because a larger distance from zero means a smaller negative number.

$-2.05 > -\frac{9}{4} > -2.5 > -\frac{11}{4}$

Sentence to look for: something like “I turned every fraction into a decimal first, then ordered by how far each one is from zero.”